Experiment-17

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Course Code: MLA0305 **Slot:**B

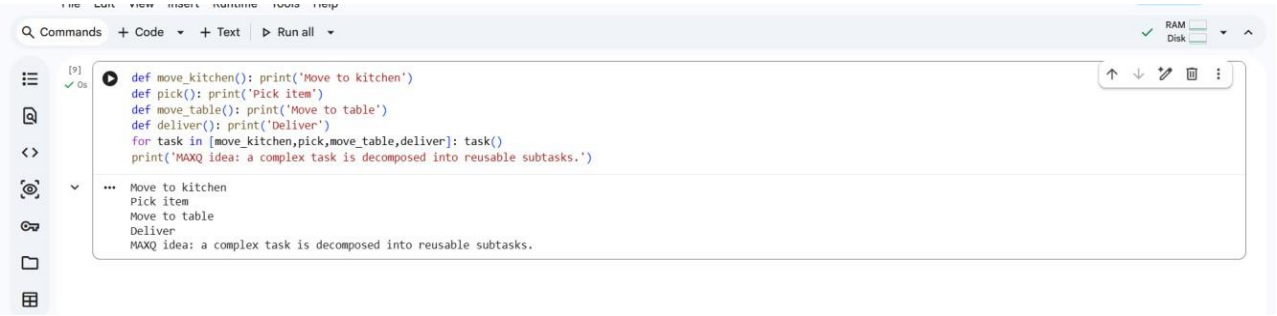
Title: Implementation of Hierarchical Reinforcement Learning Using MAXQ

Aim:

To implement Hierarchical Reinforcement Learning using the MAXQ framework and analyze hierarchical task decomposition.

Procedure:

1. Define a complex sequential decision-making problem.
2. Divide the main task into smaller subtasks.
3. Represent the subtasks using the MAXQ hierarchy.
4. Define primitive actions for lower-level tasks.
5. Define higher-level tasks that combine subtasks.
6. Train the agent at different levels of the hierarchy.
7. Execute subtasks according to the learned hierarchy.
8. Calculate cumulative rewards.
9. Compare hierarchical learning with flat RL.
10. Analyze task decomposition and learning efficiency.



Result:

The MAXQ-based hierarchical agent successfully decomposed the complex task into subtasks. Hierarchical learning improved the organization and efficiency of decision-making.